SOUND

Time: 1h **Marks:40**

SECTION I

[*All questions are compulsory*]

Question no.1 **[5]**

[i] A chord stretched between two fixed supports is plucked precisely at the centre after which it is released. Considering the resistance offered by the medium which type of vibration does it execute?

[a] Free Vibrations

[b] Resonant vibrations

[c] Damped vibrations

[d]Forced vibrations.

[ii] On the surface of moon when a bob of pendulum of length 500cm is displaced from its mean position, calculate the natural frequency of the pendulum during free vibrations.

[a] 0.15Hz

[b] 0.12Hz

[c] 0.09Hz

[d] 0.06Hz

[iii] A man stands in between two vertical cliffs X and Y such that he is at 660m from X. When he explodes the cracker, he hears the first echo after 4second and second echo 6 second later. Calculate the distance of cliff Y from the man.

[a] 1400m

[b] 1500m

[c] 1650m

[d] 1700m

[iv] A person standing between two vertical cliff whistles. He later moves towards the cliff by travelling 300m at first, he hears the echo after 4 second and then on travelling a distance hears the echo after 9second. Time taken for first echo is …. And for second echo is …..

[a] 4 s and 5 s

[b] 5s and 4 s

[c] 9s and 4 s

[d] 4s and 9s.

[v] Bats can catch their prey by using the phenomenon of….

[a] refraction of sound

[b] dispersion of sound

[c] echo

[d] reflection of sound.

Question 2: [5]

[i] Sound waves require …… [material medium / vacuum] for its propagation.

[ii] Resonance is special case of ……. [forced/ damped] vibrations.

[iii] During resonance the body starts vibrating with …… [large / maximum] amplitude producing a loud sound.

[iv] Loudness being a …… [qualitative / quantitative] aspect has …… [no unit / Phon as a unit].

[v] Frequency of stretched string is inversely proportional to square root of …… [tension / length of string / linear density of string].

Question 3:

[a] Prove that minimum distance for an echo to occur is 16.6m **[2]**

Ans: Consider the velocity of sound in air = 332ms-1

For an echo to occur, wave velocity = total distance travelled / time

$$332 = [2 \times \text{distance}] / 0.1$$

$$33.2 = 2d$$

$$d = 16.6\text{m}$$

The minimum distance for an echo to occur is 16.6m

[b] Two sound waves travelling in air have their time period in the ratio 2:3. Find the ratio of their wavelengths. **[2]**

Ans: Since the two sound waves are travelling in the same homogeneous medium their velocities would be the same.

Using the relation,

Wave velocity = frequency x wavelength,

$V_1 = f_1.\lambda_1$ and $V_2 = f_2.\lambda_2$

So, $f_1.\lambda_1 = f_2.\lambda_2$

$f_1 / f_2 = \lambda_2 / \lambda_1$

$1/T_1 / 1/T_2 = \lambda_2 / \lambda_1$

$T_2 / T_1 = \lambda_2 / \lambda_1$

$T_1 / T_2 = \lambda_1 / \lambda_2$

$2 / 3 = \lambda_1 / \lambda_2$

Hence the ratio of wavelength is 2:3

[c] Why is sound wave generally characterized as a mechanical wave? [2]

Ans: Sound wave is generally characterized as a mechanical wave as it requires material medium to propagate and such medium should possess inertia too.

[d] Why are hollow sound boxes provided for musical stringed instruments? [2]

Ans: Hollow sound boxes are provided with musical stringed instruments as larger the surface area of the vibrating body louder is the sound heard.

[e] Sound produced by church bells is greater than sound produced by bells of wind chimes. Give reason for this occurrence. [2]

Ans: Loudness of sound is directly proportional to the surface area of vibrating surface. Since church bells have larger surface area as compared to bells of wind chimes a louder sound is heard in case of church bells.

SECTION II

[*Attempt **any two** out of three questions*]

Question 4:

[a] State any four factors affecting loudness of sound? [4]

Ans: Four factors affecting loudness of sound are:

[i] Amplitude of vibrations: Loudness is directly proportional to square of the amplitude.

[ii] Surface area of vibrating body: Larger the surface area of vibrating body larger is the loudness of sound heard

[iii] Density of the medium: Greater the density of medium louder is the sound heard

[iv] Inertia of the medium: Greater the inertia of the medium greater is the loudness of sound.

b] Differentiate between free and damped vibrations [3]

Ans:

FREE VIBRATIONS	DAMPED VIBRATION
1. These vibrations occur under the absence of external periodic force	1. These vibrations occur in the presence of external periodic force
2. These vibrations have constant amplitude during its propagation	2. These vibrations have decreasing amplitude as the wave propagates through the medium
3. No loss of energy through its course of propagation	3. Continuous dissipation of energy due to frictional air resistance or resistance offered by medium.

c] A person standing between 2 vertical cliffs hears two sounds one after 3 second and another 4 second later. Find the distance between the person and the farthest cliff assuming the velocity of sound in air as $330ms^{-1}$ [3]

Ans: Since we must find the distance from the farthest cliff the time taken would be more which would be [3 + 4] second as the second echo is heard 4 second later than the first echo.

For the second echo to occur,

Wave velocity = total distance travelled by wave / time taken

330 = 2 [distance] / 7second

2d = [330 x 7]

2d = 2310

d = 1155 m

The distance of the farthest cliff from the person is 1155m.

Question 5:

[a] Name the waves used for echo depth sounding. Give a reason why such waves are used for this purpose. Also state why these waves are not audible to us? [3]

Ans: The waves used for echo depth sounding are Ultrasonic waves as they can travel undeviated through large distances and cannot be absorbed easily by the medium. These waves have frequencies above 20KHz which is above the audible range of hearing for a human ear.

b] A vibrating tuning fork is placed over the mouth of the burette filled with water. The tap of the burette is opened and water level gradual starts falling. It is found that the sound from the tuning fork becomes very loud at a particular length of the water column.

[i] Name the phenomenon taking place when this occurrence is noted.

Ans: The phenomenon taking place when a very loud sound is heard is Resonance in air column.

[ii] Sound become loud for this length of water column. Why? [3]

Ans: As water column decreases the air column inside the burette increases. It is this length of air column that is set into forced vibrations by the vibrating tuning fork placed at the mouth of the burette. When the frequency of the air column matches or is an integral multiple of the frequency of the tuning fork a loud sound is heard.

c] A man standing in front of a vertical cliff fires a gun. He hears an echo after 3 sec. On moving closer to the cliff by 83.5m he fires again and hears the echo after 2.5sec. Calculate:

[i] Distance of the cliff from original position of the man

[ii] Velocity of sound [4]

Ans: Case 1:

For the first echo to occur,

Wave velocity = [2 x distance / time]

= 2 x d / 3.....Eqn 1

Now in the second case the man moves closer to the cliff and then fires the gun and hears the second echo after 2.5second.

For the second echo,

Wave velocity = total distance travelled / time taken

= 2 [d – 83.5] / 2.5

= [2d – 167] / 2.5..... Eqn 2

Since velocity of wave remains constant in a homogeneous medium,

Eqn 1 = Eqn 2

2d / 3 = [2d – 167] / 2.5

5d = [6d – 501]

d = 501m

Hence the distance of cliff from original position of man is 501m

Wave velocity = [2 x distance] / time

= [2 x 501] / 3

= 1002 /3

= $334ms^{-1}$

Hence the velocity of sound = $334ms^{-1}$.

Question 6:

[a] State the methods of reducing noise pollution? [3]

Ans: Noise pollution can be reduced by the following methods:

[i] Using sound absorbers

[ii] Planting more trees in the area

[iii] using garden walls so that sound gets absorbed by plants.

[b] How is a radio set tuned to a particular transmitting station? Name this phenomenon **[3]**

Ans: The radio set tunes to the frequency of the transmitting station through the oscillating disc present in the tuner which leads to a loud and clear sound. The phenomenon occurring is called resonance.

c] Give reason for the following: **[4]**

[i] Soldiers are asked to break their footsteps while crossing a suspension bridge.

Ans: Soldiers are asked to break their footsteps while crossing a suspension bridge as the cumulative frequency of their footsteps may match the natural frequency of the bridge and the bridge may vibrate with maximum amplitude causing it to collapse.

[ii] Rear view mirror of the motorbike starts vibrating violently when driven at a particular speed.

Ans: Rear view of mirror starts vibrating violently when driven at a particular speed as the frequency of the piston of engine may match the natural frequency of the rear-view mirror as a result would start vibrating violently.